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Taar

A phone can tell you whether a wire is carrying current — from three centimetres away, touching nothing, with the radio off.

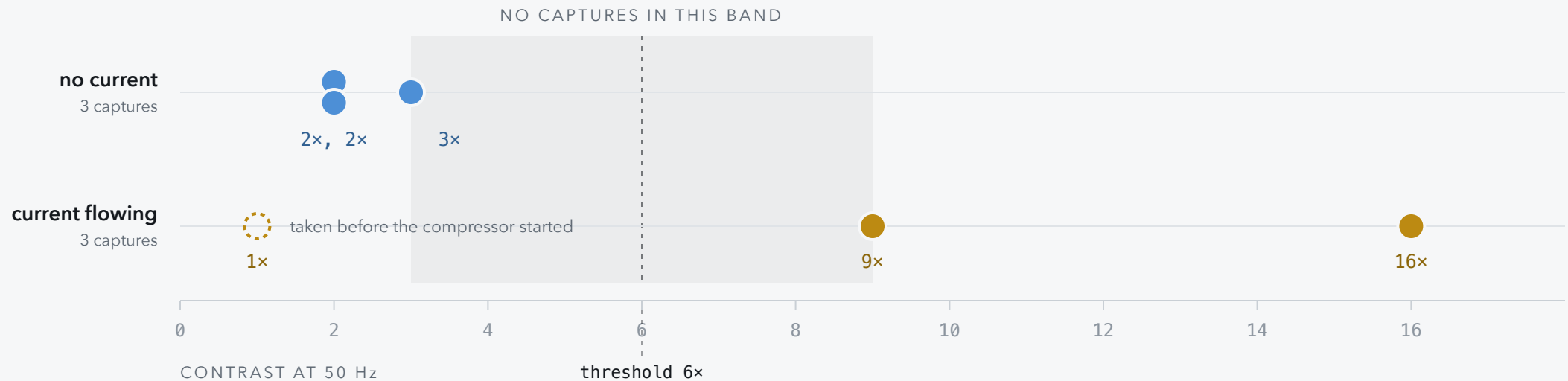
We built it and measured it. **Six captures against a live supply cable: no current read 2-3x, current flowing read 9-16x.**

Goutham Pakala · Geetha Tigulla

Phase 1 idea submission

44/44 checks · 9.3 MB APK · 0 network permissions

Two populations, and a gap between them.



A factor of three between the populations, no overlap. The threshold moved to 6x because of this plot – the previous value, chosen from simulation, sat close enough to the idle readings that a dead circuit could read as live.

One capture in the lower lane read 1x. It was taken in the same minute as the last idle reading, before the compressor had started, and a fridge draws almost nothing until it does. **We set it aside on an explanation formed after the fact, and the separation depends on that being right.**

Same phone. Same cable. One switch.

Two pairs, taken minutes apart. The phone was taped down and not touched; the only thing that changed in the room was a switch at the wall.

APPLIANCE OFF · NOTHING MOVED

Mains detected	3x contrast · 0.27
Field amplitude	0.05 μ T

Mains detected	2x contrast · 0.15
Field amplitude	0.05 μ T

COMPRESSOR RUNNING · SAME POSITION

Mains detected	16x contrast · 0.64
Field amplitude	0.14 μ T

Mains detected	9x contrast · 0.50
Field amplitude	0.10 μ T

Both numbers come off the device, not from a model. Contrast 2–3× with no current; 9–16× with the compressor drawing. Field amplitude 0.05 μ T against 0.10–0.14, on a noise floor we measured at 0.40.

What this does **not** show is a sensitivity figure. One appliance, roughly 1–1.5 A, through a cable whose live and neutral largely cancel – enough to establish the effect, not to quantify it.

Two physical facts nobody has pointed a phone at.

$$B = \mu_0 I / 2\pi r$$

A conductor broadcasts its current

At three centimetres, five amperes produce about 33 μT . Earth's field is 45 μT and the sensor floor is 0.40. The signal is not small – it is the same order as the planet.

This is the principle a clamp meter uses. Taar does it without the clamp, which is the whole product.

$$100 \text{ Hz} = 2 \times \text{mains}$$

An arc broadcasts its fault

A series arc re-ignites twice per cycle, so its broadband noise is amplitude-modulated at 100 Hz. We look for energy **at that frequency**, not for energy.

Against four confounders – a ballast hum, speech, a sleeve brushing the phone, a motor – a plain energy detector fired on every one, every time. Ours held its 5% design point.

Both run on the device, and that is not a scoring choice. Distribution boards live in basements with no signal. A building's circuit layout is not a contractor's data to upload. And a measurement of response time cannot have a network round trip inside it.

WHO IT IS FOR

The instruments cost more than the firm earns in a month.

A clamp meter, a thermal camera and an arc-fault analyser are what make a loose connection visible before it starts a fire. For a two-person electrical firm that is a capital outlay they cannot justify, so the tools stay with large contractors and the buildings that most need checking never get checked.

Every one of those electricians already carries a magnetometer and a microphone. Taar is the argument that the instrument is already in their pocket.

Before touching a circuit, is it dead?

probe it, or trust the label

Is this circuit over its rating?

clamp meter

Is anything arcing behind the plate?

arc-fault analyser

Cost of all three

out of reach

Taar answers the first two from a three-second capture, and attempts the third. It reports a relative index, not amperes, unless the circuit has been calibrated against a known load.

Everything we got wrong, we got wrong in simulation.

The app passed every check we had written and still returned nothing on a real phone. These four only appeared once it was installed.

01	Two Android clocks	<code>SensorEvent.timestamp</code> counts deep sleep; <code>System.nanoTime()</code> does not. Comparing them ended every capture on its first sample.
02	Magnitude destroyed the signal	Reducing three axes to $\sqrt{x^2+y^2+z^2}$ hid the AC component behind Earth's static field. A 1 μ T signal moved it by 0.01.
03	Normalised by the wrong quantity	A hand-held capture drifts several μ T. That drift buried the 50 Hz term and reported a real signal as zero confidence.
04	A threshold we guessed	The arc floor was five times too small, and called a domestic fridge an arcing fault. Re-set from nine measured captures.

A simulation that had predicted any of these would have been a better simulation. It did not – which is the case for building early and putting it on the hardware, and the reason we would rather show this list than claim the first build worked.

Not a concept.

Pre-check

Measured, not requested. A reading is only as good as what the sensors actually delivered.

OK

Magnetometer

present

OK

Sample rate

105.3 Hz measured · requested 99 Hz · 21 distinct phases

OK

Timing jitter

0.0% of the interval — uniform, but 21 phases is plenty

OK

Noise floor

0.40 μ T RMS

OK

Microphone

fell back to MIC — arc band may be attenuated

!!

Reference

none — record a baseline before measuring

PRE-CHECK · MEASURED, NOT REQUESTED

Logic checks passing	44 / 44
Debug APK, builds clean	9.3 MB
Network permissions in the manifest	0
Magnetometer rate achieved	105.3 Hz · 21 phases
Noise floor, two sessions	0.39 – 0.40 μ T
Faults in the catalogue, English & Telugu	6

Signal path

Per-axis sine fit at 50 Hz, detrended, amplitude as the vector sum. Lomb-Scargle contrast against 30–48 and 52–70 Hz. Scoring by median and MAD against the circuit's own reference.

On top of it

A rules table that shows its evidence. A nearest-centroid classifier **trained on the device** from the technician's labels, which rejects rather than guessing. No language model is built; one is planned for the build window.

What Taar does not do.

It is a triage aid, not a certification

It does not replace a licensed electrician, a calibrated clamp meter, an insulation tester or a statutory inspection. The app says so on the result screen, not in a settings page.

The acoustic half is designed, not validated

No real arc has been recorded. The magnetometer half rests on textbook physics and now on measurement; the arc half rests on neither yet, and we will say so on stage before a judge asks.

It reports an index, not amperes

Unless the circuit has been calibrated against a known load. A fabricated ampere figure is worse than no figure.

It refuses rather than guesses

With no reference, or an ill-conditioned fit, it returns a refusal. An electrician acting on a confident wrong reading is the worst outcome this project has, and the code is arranged so that outcome is unreachable.

Our pre-event work is a research spike on a separate branch, rebuilt from an empty scaffold inside the event window and never merged. The protocol is written down, and we will confirm it with an organiser at check-in.

NINETY SECONDS, ON THE IQOO, IN AIRPLANE MODE

A phone finds electricity through insulation, and touches nothing to do it.

01

Off. Phone taped to the cable. Measure. No mains detected.

02

On. Nothing in the room moves but a switch. Measure again.

03

The jump. The contrast number is the demo. Everything else is set-up.

Yesterday it was a claim.

Today there are six captures behind it.

Taar · Against Default
Goutham Pakala · Geetha Tigulla